General Probability Review

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1 Slide Deck 1: Counting and Combinatorics

1.1 The Basic Principle of Counting

! Basic Principle of Counting

Suppose a process has two stages. If the first stage has m possible outcomes, and for each outcome of the first stage the second stage has n possible outcomes, then the entire process has

$$\boxed{mn}$$

possible outcomes.

The important idea is to think of the process as occurring in **stages**. If each first-stage choice leads to n completed outcomes, then

$$\underbrace{n + n + \cdots + n}_{m \text{ times}} = mn.$$

! Generalized Basic Principle of Counting

Suppose a process consists of r stages. If there are

$$n_1, n_2, \ldots, n_r$$

choices at the successive stages, then the total number of possible outcomes is

$$\boxed{n_1 n_2 \cdots n_r}.$$

Useful consequence: number of subsets

A set containing n distinct objects has

$$\boxed{2^n}$$

different subsets, including the empty set and the entire set.
For each object, there are two choices: include it or do not include it.

1.2 Permutations

! Permutations

A **permutation** is an ordered arrangement of distinct objects.
The number of ways to arrange n distinct objects in order is

$$\boxed{n! = n(n-1)(n-2)\cdots 2 \cdot 1}.$$

Thus, $n!$ can be interpreted as the number of permutations of n distinct objects.

Key question: Does order matter?

If changing the order creates a different outcome, think about permutations or the multiplication principle.

1.3 Combinations

! Combinations

A **combination** is an unordered group.
The number of ways to choose r objects from n distinct objects is

$$\boxed{\binom{n}{r} = \frac{n!}{r!(n-r)!}.}$$

Notation: $\binom{n}{r}$ is read "n choose r."
By convention,

$$\boxed{\binom{n}{0} = 1}.$$

Why do we divide by $r!$? The ordered count

$$\frac{n!}{(n-r)!}$$

counts each group of r people once for every possible ordering of those r people. Since there are $r!$ such orderings, we divide by $r!$.

1.4 Binomial and Multinomial Coefficients

Binomial Theorem

$$\boxed{(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}.}$$

The coefficient $\binom{n}{r}$ appears because to obtain $x^r y^{n-r}$, we must choose exactly r of the n factors to contribute an x.

Multinomial Coefficient

If

$$n_1 + n_2 + \cdots + n_r = n,$$

then

$$\boxed{\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}.}$$

! Multinomial Interpretation 1: Labeled Groups

The multinomial coefficient

$$\binom{n}{n_1, n_2, \ldots, n_r}$$

is the number of ways to divide n **distinct** objects into r **distinct/labeled** groups of sizes

$$n_1, n_2, \ldots, n_r.$$

Equivalently,

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots = \frac{n!}{n_1!\cdots n_r!}.$$

Because the groups are labeled, we do *not* divide by $r!$.

! Multinomial Interpretation 2: Repeated Objects

The multinomial coefficient

$$\binom{n}{n_1, n_2, \ldots, n_r}$$

also counts the number of ordered arrangements of n objects when

$$n_1 \text{ are of type 1}, \quad n_2 \text{ are of type 2}, \quad \ldots, \quad n_r \text{ are of type } r.$$

Thus,

$$\boxed{\text{number of arrangements} = \frac{n!}{n_1!n_2!\cdots n_r!}.}$$

The factorials in the denominator remove the overcounting caused by indistinguishable objects of the same type.

Multinomial Theorem

$$\boxed{(x_1 + \cdots + x_r)^n = \sum_{\substack{n_1+\cdots+n_r=n \\ n_i \geq 0}} \binom{n}{n_1, \ldots, n_r} x_1^{n_1} \cdots x_r^{n_r}.}$$

1.5 Counting: Quick Decision Guide

Situation	**Useful count**
Several successive choices	Multiply the numbers of choices
Arrange all n distinct objects	$n!$
Choose r objects from n; order does not matter	$\binom{n}{r}$
Split n distinct objects into labeled groups of prescribed sizes	$\binom{n}{n_1,\ldots,n_r}$
Arrange objects with repeated types	$\dfrac{n!}{n_1!\cdots n_r!}$

2 Slide Deck 2: Sample Spaces, Events, and Event Algebra

2.1 Sample Spaces

! Sample Space

The **sample space** is the set of **all possible outcomes** of an experiment.
Notation:

$$\boxed{S}.$$

Important conceptual point

The sample space depends on **what information we record**.
For example, when flipping a coin twice:

$$S = \{HH, HT, TH, TT\}$$

if we record the result of each toss, while

$$S = \{0, 1, 2\}$$

if we record only the number of heads.

2.2 Events

! Event

An **event** is any subset of the sample space S.
If E is an event, saying that "E occurs" means that the outcome of the experiment belongs to E.

2.3 Union

! Union

For events E and F,

$$E \cup F$$

is the event containing all outcomes that are in E, in F, or in both.
In words:

$$\boxed{E \cup F = \text{“}E \text{ or } F \text{ occurs”}}.$$

Here "or" is inclusive: both events are allowed to occur.

2.4 Intersection, Null Events, and Disjoint Events

! Intersection

For events E and F,

$$E \cap F$$

is the event containing outcomes that belong to **both** E and F.
In words:

$$\boxed{E \cap F = \text{“}E \text{ and } F \text{ occur”}}.$$

Null Event and Disjoint Events

The **null event** or **empty event** is

$$\boxed{\emptyset},$$

the event that cannot occur.
Events E and F are **mutually exclusive** or **disjoint** when

$$\boxed{E \cap F = \emptyset}.$$

In other words, they cannot occur at the same time.

2.5 Finite and Countable Unions and Intersections

For events $E_1, \ldots, E_n$,

$$\bigcup_{j=1}^{n} E_j = \{\text{at least one of the events occurs}\},$$

while

$$\bigcap_{j=1}^{n} E_j = \{\text{all of the events occur}\}.$$

Similarly, for a countable collection $E_1, E_2, \ldots$,

$$\bigcup_{n=1}^{\infty} E_n = \{\text{at least one } E_n \text{ occurs}\},$$

and

$$\bigcap_{n=1}^{\infty} E_n = \{\text{every } E_n \text{ occurs}\}.$$

2.6 Complements and Containment

! Complement

The **complement** of E is the event containing all outcomes in S that are not in E.
Notation:

$$\boxed{E^c}.$$

Thus,

$$E^c = \{E \text{ does not occur}\}.$$

Also,

$$S^c = \emptyset.$$

Containment

If every outcome in E is also an outcome in F, then

$$\boxed{E \subseteq F}.$$

If $E \subseteq F$, then whenever E occurs, F must also occur.
Two events are equal precisely when

$$E \subseteq F \qquad \text{and} \qquad F \subseteq E.$$

2.7 Laws of Event Algebra

Commutative Laws

$$E \cup F = F \cup E, \qquad E \cap F = F \cap E.$$

Associative Laws

$$(E \cup F) \cup G = E \cup (F \cup G),$$
$$(E \cap F) \cap G = E \cap (F \cap G).$$

! Distributive Laws

$$\boxed{(E \cup F) \cap G = (E \cap G) \cup (F \cap G)}$$

and

$$\boxed{(E \cap F) \cup G = (E \cup G) \cap (F \cup G).}$$

! De Morgan's Laws

For events $E_1, \ldots, E_k$,

$$\boxed{\left(\bigcup_{n=1}^{k} E_n\right)^c = \bigcap_{n=1}^{k} E_n^c}$$

and

$$\boxed{\left(\bigcap_{n=1}^{k} E_n\right)^c = \bigcup_{n=1}^{k} E_n^c.}$$

For two events in particular,

$$(E \cup F)^c = E^c \cap F^c,$$
$$(E \cap F)^c = E^c \cup F^c.$$

Mental Translations

$$\begin{aligned} E \cup F &: \text{ at least one occurs,} \\ E \cap F &: \text{ both occur,} \\ E^c &: E \text{ does not occur.} \end{aligned}$$

3 Slide Deck 3: Probability Axioms and Their Consequences

3.1 The Probability Axioms

A probability model assigns a number $\mathbb{P}(E)$ to each event E.

! The Three Probability Axioms

Axiom 1: For every event E,

$$\boxed{0 \leq \mathbb{P}(E) \leq 1}.$$

Axiom 2:

$$\boxed{\mathbb{P}(S) = 1}.$$

Axiom 3: If $E_1, E_2, \ldots$ are mutually exclusive, then

$$\boxed{\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i).}$$

These axioms must be satisfied by any valid assignment of probabilities.

3.2 Consequences of the Axioms

Consequence 1

$$\boxed{\mathbb{P}(\emptyset) = 0}.$$

Consequence 2: Finite Additivity

If $E_1, \ldots, E_n$ are mutually exclusive, then

$$\boxed{\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} \mathbb{P}(E_i).}$$

In particular, if A and B are disjoint,

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).}$$

3.3 Equally Likely Outcomes

! Equally Likely Outcomes

Suppose

$$S = \{x_1, \ldots, x_N\}$$

and every outcome is equally likely, so

$$\mathbb{P}(\{x_i\}) = \frac{1}{N}.$$

Then for any event E,

$$\boxed{\mathbb{P}(E) = \frac{\text{number of outcomes in } E}{\text{number of outcomes in } S} = \frac{|E|}{|S|}.}$$

This is where combinatorial counting is especially useful.

Probability Space

A triple

$$(S, \mathcal{F}, \mathbb{P}),$$

where S is the sample space, $\mathcal{F}$ is the collection of events, and $\mathbb{P}$ satisfies the three probability axioms, is called a **probability space** or **probability model**.

3.4 General Propositions

Proposition 1: Complement Rule

For every event E,

$$\boxed{\mathbb{P}(E^c) = 1 - \mathbb{P}(E).}$$

This follows from

$$E \cup E^c = S \qquad \text{and} \qquad E \cap E^c = \emptyset.$$

Proposition 2: Monotonicity

If

$$E \subseteq F,$$

then

$$\boxed{\mathbb{P}(E) \leq \mathbb{P}(F).}$$

A smaller event cannot have greater probability than an event containing it.

Proposition 3: Addition Rule

For any two events E and F,

$$\boxed{\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F).}$$

The subtraction is necessary because the outcomes in $E \cap F$ were counted twice when $\mathbb{P}(E)$ and $\mathbb{P}(F)$ were added.

3.5 Inclusion–Exclusion for Three Events

For three events,

$$\boxed{\begin{aligned}\mathbb{P}(E \cup F \cup G) = \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) \\ - \mathbb{P}(E \cap F) - \mathbb{P}(E \cap G) - \mathbb{P}(F \cap G) \\ + \mathbb{P}(E \cap F \cap G).\end{aligned}}$$

Why is the triple intersection added back?

An outcome in all three events is

- counted three times in $\mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G)$;
- subtracted three times in the pairwise-intersection terms.

At that point it has been counted zero times, so we add the triple intersection once.

Useful Strategy: Use the Complement

When an event such as "at least one" is hard to count directly, its complement may be much easier. For example,

$$\boxed{\mathbb{P}(\text{at least one}) = 1 - \mathbb{P}(\text{none}).}$$

Similarly,

$$\mathbb{P}(\text{at least one repeated birthday}) = 1 - \mathbb{P}(\text{all birthdays different}).$$

4 Slide Deck 4: Conditional Probability

4.1 Conditional Probability

The conditional probability of E given F asks for the probability that E occurs once we know that F has occurred.

Notation:

$$\boxed{\mathbb{P}(E \mid F)}.$$

Definition of Conditional Probability

If

$$\mathbb{P}(F) > 0,$$

then

$$\boxed{\mathbb{P}(E \mid F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}.}$$

Reduced Sample Space Interpretation

Conditioning on F means that F becomes the new reference set.
Instead of asking

"What fraction of the whole sample space lies in E?"

we ask

"What fraction of the outcomes in F also lie in E?"

The numerator is the part of F in which E also happens:

$$E \cap F.$$

4.2 Conditional Probability with Equally Likely Outcomes

If the original outcomes are equally likely, then

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}$$

becomes

$$\boxed{\mathbb{P}(E \mid F) = \frac{|E \cap F|}{|F|}.}$$

Thus, when conditioning on F, we can often think of F itself as the new sample space.

Important: the denominator is now the number of outcomes in F, not the number of outcomes in the original sample space.

4.3 Multiplication Rule

! Computing Intersections by Conditioning

From the definition of conditional probability,

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}.$$

Rearranging gives

$$\boxed{\mathbb{P}(E \cap F) = \mathbb{P}(F)\mathbb{P}(E \mid F).}$$

! General Multiplication Rule

For events $E_1, E_2, \ldots, E_n$,

$$\boxed{\begin{aligned}\mathbb{P}(E_1 \cap E_2 \cap \cdots \cap E_n) = \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1) \\ \times \mathbb{P}(E_3 \mid E_1 \cap E_2) \cdots \\ \times \mathbb{P}(E_n \mid E_1 \cap \cdots \cap E_{n-1}).\end{aligned}}$$

For three events,

$$\boxed{\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1)\mathbb{P}(E_3 \mid E_1 \cap E_2).}$$

The idea is sequential:

1. E_1 occurs;
2. then E_2 occurs given that E_1 already occurred;
3. then E_3 occurs given that E_1 and E_2 already occurred;
4. continue in the same way.

5 One-Page Formula Checklist

Counting

$$\boxed{\text{successive stages: } n_1 n_2 \cdots n_r}$$

$$\boxed{\text{permutations: } n!}$$

$$\boxed{\text{combinations: } \binom{n}{r} = \frac{n!}{r!(n-r)!}}$$

$$\boxed{\text{multinomial: } \binom{n}{n_1, \ldots, n_r} = \frac{n!}{n_1! \cdots n_r!}}$$

Event Algebra

$$\boxed{(E \cup F)^c = E^c \cap F^c} \qquad \boxed{(E \cap F)^c = E^c \cup F^c}$$

$$\boxed{(E \cup F) \cap G = (E \cap G) \cup (F \cap G)}$$

$$\boxed{(E \cap F) \cup G = (E \cup G) \cap (F \cup G)}$$

Probability

$$\boxed{\mathbb{P}(S) = 1}, \qquad \boxed{\mathbb{P}(\emptyset) = 0}, \qquad \boxed{\mathbb{P}(E^c) = 1 - \mathbb{P}(E)}$$

If $E \subseteq F$,

$$\boxed{\mathbb{P}(E) \leq \mathbb{P}(F)}.$$

$$\boxed{\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)}$$

For equally likely outcomes,

$$\boxed{\mathbb{P}(E) = \frac{|E|}{|S|}}.$$

Conditional Probability

$$\boxed{\mathbb{P}(E \mid F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}}$$

For equally likely outcomes,

$$\boxed{\mathbb{P}(E \mid F) = \frac{|E \cap F|}{|F|}}.$$

$$\boxed{\mathbb{P}(E \cap F) = \mathbb{P}(F)\mathbb{P}(E \mid F).}$$

Before calculating: identify the sample space, identify the event, and decide what is actually being counted.